

Abstract

The aim of this sub-project is to develop an image processing component to support low-cost automated greenhouses. The image processing application must enable the greenhouse system to establish various aspects about the plants from the images taken. These aspects include plant health, plant growth and plant disease identification from the images of different parts of the plants in the greenhouse.

What is the purpose of this section?

- Context and importance of your research.
- Outline of literatur review

Why choose image processing techniques?

- Former methods cost too much time and were expensive + destructive [11]
- Focus on image analysis component [4]

1 Plant Phenotyping

Introduction Plant breeding is defined as the process of inducing desirable inheritable traits in plants through controlled human action [18]. This process is important in solving the problem of adequate global food supply. The rapid growth of the human population has created a strain on global food supply. It is estimated that crop production must increase by 70% by 2050. [3] to meet this demand. Additionally, abiotic and biotic stressors limit cultivar biomass and yield [6]. Therefore, plant breeders need effective techniques to produce germplasm for cultivars that maximise yield and biomass, in various adverse environmental conditions. In the last few decades, image processing techniques in plant phenotyping have been shown to be an effective tool for plant breeding [1].

What Is Plant Phenotyping?

Plant phenotyping is the evaluation of phenotypes in plants [5]. Phenotypes are primitive or complex traits of an organism that can be observed or quantified and evaluated numerically [9]. Recent advances in plant phenotyping measurement indicate that a phenotype can alternatively be seen as combination of all morphological, physiological, chemical, and behavioral aspects that represent the individual plant [7, 17, 14].

Plant Phenotyping As Method For Studing Plant-Environment Interactions In Plant Breeding

Discuss a typical workflow

A Taxonomy Of Phenotypes: What is currently phenotypically measurable?

Discuss various phenotypes traits that can be measured, and relate them to the goal of plant breeding.

The Relationship Between Image Processing And Plant Phenotyping

Discuss why and how image processing is done with plant phenotyping

- Examples of how plant phenotyping has been used with image processing (in field and in greenhouse growth chambers)
- Therefore image processing combined with plant phenotyping provide the fundamental tools that allow us to automatically analyse greenhouse cultivar performance

The field of phenomics and in particular the methodologies of plant phenotyping fill this gap, and have been applied successfully to measure, analyse and identify various traits in plants that predict yield, stress and biomass using image processing techniques. Plant phenotyping forms the basis of modern image processing techniques for the management of greenhouses

Explanation of relevance of plant phenotyping for current study? The term phenotype was first employed by Johannsen [9]. Who stated that a phenotype is all types of organism which are identified from one another by inspection or by some measurement. So it is an observable primitive trait or a complex trait of a plant that can be identified visually or measured and evaluated numerically. Therefore the data produced by plant phenotyping assist in choosing the genetic material (germplasm) that will enable plant breeders to induce traits in cultivars which are optimal for various different growing environments (like water scarce environments).

Why plant phenotyping in the context of this study? Conventional plant phenotyping suffers from scalability issues and is often constrained by costs, time, and human effort required. Furthermore many methods are destructive to the cultivar, making it hard to replicate, and compare measurements of phenotypes over time through the ontogenesis of a cultivar. However in the past decade image processing techniques have been applied successfully to acquire phenotypical data in a non-destructive manner [2]. There exist many different image techniques that have been applied in greenhouses and in the field which allow for the assessment of resistance to disease, biomass, soil salinity, etc using various different sensors, and techniques.

How can we measure plant stresses and disturbances?

Stresses and disturbances of plants are induced by environmental conditions the organisms are placed in, and therefore transitively induce phenotypical traits. We can exploit this fact by measuring plant phenotypical traits, we can infer the presence of potential stresses and disturbances on cultivars. For example thermal imaging allows to measure the phenotype of plant temperature or plant transpiration, which in turn can give us an indication of soil salinity. We indirectly infer environmental conditions based on phenotypical traits

What physical traits of a plant can be measured phenotypically?

Conceptual basis Typically in the context of plant breeding we care about plant yield, resistant to abiotic and biotic stresses. That is in general we seek plants which are fit. In the literature of the term trait is used to describe individual scoped properties, community scoped properties and even environmental properties. trait is any morphological, physiological or phenological feature measurable at the individual level, from the cell to the whole-organism level, without reference to the environment [27]. Plant phenotyping reflects this in its achievements and developments over the past decade. The key is environmental related performance metrics are derived from individual plant traits. So these phenotypical traits act as proxies that indirectly predict plant fitness and performance.

Morphological traits About plant sturcture, which affects its performance and gives indication regarding its health

- Leaf width
- Number of leaves
- Leaf inclination angle
- Leaf area
- Root system architecture (e.g root diameter)
- Plant shoots
- Fruit
- Flowers

Physiological traits About how plant operates in repsonse to its growing conditions

- Stomotal conductance
- Chlorophyll content of leaf
- Plant water status
- Plant photosynthesis (metablism)
- Nutrient content

Phenological traits These are phenological traits (when certain biological events occur that depend on climates)

- When fruit or flowers are yielded
- Yield prediction

The use of plant phenotyping and detecting plant stress Plant biomass is affected by stresses and disturbances [6]. A stressor is anything which stymies the growth of plant biomass. Examples of stressors include things like water scarcity, inadequate amounts of light, and unfavourable temperatures. A disturbance is anything which causes some level of destruction to the plant. Examples of disturbances include pathogens, fires, frost, and pests

Biotic Stress + Disturbance.

- Pathogenic organisms
- Pests
- Parasitic plants

Abiotic Stress + Disturbance

- Drought
- Flood
- Salinity
- Heat Stress
- Raditation Stress, light stress

What is the purpose of this section?

- Discuss image processing in the context of plant phenotyping
- Discuss image acquisition techniques
- Discuss image processing techniques used in plant phenotyping

How does image processing relate to plant phenotyping?

Fill out this section

The conceptual pipeline for image processing techniques in plant phenotyping

This paragraph should discuss how image processing is applied. I.e it should map the process from image acquisition to image processing to data generation/analysis

Foundations of image acquisition in plant phenotyping

- measure a phenotype quantitatively through the interaction between light and plants such as reflected photons, absorbed photons, or transmitted photons. Each component of plant cells and tissues has wavelength-specific absorbance, reflectance, and transmittance properties [14]
- For example, chlorophyll absorbs photons primarily in the blue and red spectral region of visible light [14]
- Imaging at different wavelengths is used for different aspects of plant phenotyping [14].
- Imaging techniques are very helpful for detecting these properties, especially for those that cannot be seen by the naked eye [14]

Image acquisition methodologies in plant phenotyping

Thermal imaging sensors

- Conceptual basis of this technique [20]
- Have been used to measure plant water status.
- Used in Irrigation scheduling.
- By inferring transpiration rate of leaves which is governed by stomatal conductance.
- Stomatal conductance can indicate drought, pathogens, and also soil salinity.
- Have also been used for yield predictions [8]

Hyperspectral sensors

- Based on the principles of spectroscopy. Can detect biochemical, morphological and physiological traits of plants. Has been used to detect drought stress. Also used to detect water stress, and pathogens. Different reflectance can tell us about different things. Certain band tells us about water stress, some can tell us about chlorophyll content. [23].
- Conceptual basis and applications [22]
- Crop yield
- Biotic and abiotic factors

Tomography

- three-dimensional visualization of plant internal structures, root systems, and physiological processes
- Based on principles of x-rays, moving sensor takes planar cross sections to 3D visualize plant
- Drought stress [26]
- Yield Estimations
- has references for usages [19]

Fluorescence sensors

- Conceptual basis of this technique [24]
- Predict metabolism using photosynthesis which tells us about growth status
- Any stress or disturbance can indirectly or directly affect plant metabolism, so chlorophyll content can be used as a measure to detect these things like pathogens, light shortage, nitrogen deficiency, water stress etc [15]
- In optimal conditions, very little heat is emitted in photosynthesis so in certain near infrared emission should be low
- Actinic light involved here?

RGB Camera

- Mimic human vision
- measure plant shoot biomass (yield potential) [14]
- leaf senescence (tissue resistance to salt)

What is the purpose of this section?

- Discuss how image processing and plant phenotyping are applied in low-cost automated greenhouses
- What methods, limitations and advances have there been.
- What does a typical image preprocessing pipeline look like?

Basis for future in low-cost imaging analysis

The basis for the points below is from [nabwireReviewApplicationArtificial2021]

- Existing phenotyping systems are commercially developed and under patent, so are not modifiable or readily available [12]
- RGB, and hyperspectral sensors most commonly used as they are affordable
- Data collection and analysis tools are not generalizable or standardised, so many custom flows are made
- Indicating the need for standardised, extendable image analysis pipelines using readily available open-source software and hardware
- Recent advances show the use of cheap hardware like raspberry pi, and affordable RGB sensors [25]
- However these sensors don't have a lot of compute power, so analysis and preprocessing is done remotely
- Cheap sensors but are cheaper and faster
- Cloud architectures exist, but with amount of data generated this cost can be prohibitive in terms of processing cost [bauckhageDataMiningPattern2013]. Cyberinfrastructure expensive ref (95,96)
- Moreover, existing systems are not designed for multiple genotypes so their applicability is limited, suggesting need for more generalizable image analysis
- Deep learning and machine learning can be used to automate image analysis, and reduce costs
- Availability of large labelled data is bottleneck
- To use on cheap hardware you also need to optimize
- Therefore low complexity efficient algorithms for automated image preprocessing and image analysis is needed
- Using multimodal segmentation is also another angle

- ref[36], one of the advantages of using ML approaches in plant phenotyping is their ability to search large datasets and discover patterns by simultaneously looking at a combination of features (compared to analyzing each feature separately). This was previously a challenge because of the high dimensionality of plant images and their large quantity, making them difficult to analyze through traditional processing methods (ref 37)
- machine learning achieved using supervised learning, so we aren't able to detect novel or unexpected traits using unsupervised learning (ref 40)
- traditional methods in image processing not generalizable, in pathogen detection and require strict conditions and custom workflows to work [2]
- Current research Focus on image sensing, rather on computation which is new bottleneck. Imaging technology has advanced, but there is lack of robust analysis and computation tools. [bauckhageDataMiningPattern2013]
- Sensors generate huge amount of data, therefore algorithms need to be designed to deal with Terabytes of data. Furthermore, low cost computing units have limited bandwidth, battery and sensor capabilities indicating that algorithms need to take this into account. Low complexity and efficient algorithms that can run on resource constrained tech is needed [bauckhageDataMiningPattern2013]
- Mobile applications can be used and run processing without need for external network, cyberinfrastructure. However there are not many applications available for this [muller-linowPlantScreenMobile2019]. Image analysis tools are often too expensive to run on these devices due to the constrained resources, however they are much cheaper and therefore can be exploited. In particular low complexity efficient systems, which generalize to various genotypes, and phenotypical measurements are needed. In particular the development of efficient low resource intensive machine modelling and deep learning techniques can be made.
- preprocessing technique choice is also needed, for example image segmentation methods can be costly, and a suitable method depends on various factors. Certain methods have cheaper cost, but less accuracy [muller-linowPlantScreenMobile2019]. In developing low-cost image analysis and preprocessing tools for phenotyping it is necessary to weight the cost savings versus accuracy and precision. But bespoke image processing techniques don't lend well to generalizability and extendability.

Current applications

- Often combined with machine learning algorithms
- Cheap Raspberry PI with RGB camera [25]
- Autonomous robots with sensors and cameras [21]
- use off the shelf commercial software like microsoft xbox360 kinect [13]

What affects a low-cost automated greenhouse cost

Points from this paper [10]

- Image preprocessing pipeline
- open source software
- hardware
- amount of labour and time needed for analysis
- energy cost of technology
- amount of data generated, where is it stored, how frequently? [4]

Challenges of low-cost automated greenhouses

- Many phenotyping systems existing are commercially developed under patent so hardware and software can not be modified. [12]
- Current applications use moving conveyor belt (benchtop) [12]
- or conveyer belt [12]
- or moving robotic arm to move sensors around to get high-throughput measurements.
- Robot solutions not cheap for farmers [21]
- large amounts of data need to be collected, and processed. Existing phenotyping platforms exist, but they are expensive and have restricted running environments.
- Many systems developed for laboratory systems, but not extended for real applications. Standardised off the shelf equipment, open-source software allow modular and extensible plus affordable solutions [16]
- Custom data collection schemes need to be made very little standards
- lack of robust methods to analyse plant images
- low cost moving track systems limited in phenotypical, genotype and management types are supported [28] suggesting need for low-cost systems that can support a wider variety of genotypes and phenotypical traits.

Fill out the conclusion section.

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